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Energy nexus for grapes production: A case study of Sangli region in India

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ABSTRACT

India is the world's ninth-largest grape-producing country. Maharashtra is a major producer of grapes and raisins, with the Sangli district accounting for a large portion of the total, hence becoming the area of focus of this research. Sangli becomes an important location for grape production. The objective of this research is to conduct a thorough, organized, and comprehensive energy analysis of the grape-growing process, emphasizing the energy consumed in important farming operations. Energy analysis is a tool used to quantify the various energies involved in the production of grapes. The share and percentage distribution of various sources of energy are determined using energy analysis. The input data needed for energy analysis is gathered from the region's major farms. Diesel (29%), electricity (22%), human labor (15%), chemical fertilizer (12%), pesticide (8%), farm yard manure (7%), machinery (6%), and irrigation (1%) are listed in decreasing order of energy usage. The energy ratio and productivity ratios are calculated as a result of the energy analysis; the corresponding values are 5.92 and 0.56 kg/MJ, respectively.

1. Introduction

The law of conservation of energy, states that energy is always conserved in a process. In all processes, energy transfer takes place from one form to another. The world is experiencing an energy crisis. Energy saved is energy conserved, so there is a clear need to address the energy crisis. There are two aspects to energy conservation: quality and cost. Quality necessitates that the product is in the greatest possible condition, and the cost reflects the amount of money spent on it. As a result, research work is required under the National Doctoral Fellowship (NDF) scheme in an indicative thrust area of "Energy Efficiency, Renewable and Sustainable Energy" as identified by the All-India Council for Technical Education (AICTE). Grapes are a subtropical fruit, and India ranks 9th in the world as a grape-growing country, with 111.4 million hectares of cultivated land generating 1234.9 million Metric Tonnes (Mt) of grapes [1]. Grape consumption has increased from 1.31 kg per capita to 3.40 kg per capita over the world (from 1970 to 2003). However, with a total grape production of around 2.58 million Mt, annual per capita availability in India is 2 kg. With this massive figure, and assuming a 4% growth rate, India alone will be able to supply 9.75 million Mt of grapes to the world by 2050, as part of the *"Feed the World by 2050"* mission [2].

With a massive yearly grape production, Maharashtra is India's biggest grape producer, accounting for around 80% of total grape production (Table 1). Grapes are grown on around 60% of the state's total agricultural land. Sangli, Nashik, Pune, Solapur, and Osmanabad are the top grape-growing districts in Maharashtra. The largest grape-producing districts are listed by state, to emphasize Maharashtra's dominance as the country's leading grape-producing state (Table 2). Agriculture is one of Sangli's most common occupations. To add to that, grape production accounts for 30% of Sangli's entire agricultural area [3]. It is because Sangli has favorable climatic conditions for grape production with January to March being the grape harvesting season. In the year 2017, out of 502.54 Mt of grape production in the Sangli district, around 97% (486 Mt) were exported to the market. Netherland, Bangladesh, United Arab Emirates, etc. are some major grapes importing countries from India [4]. India's percentage share of agricultural export rises to 12.26% of the total commodity exports in the last five years [5]. Sangli is primarily populated by small-scale grape growers with a smallholding. Nonetheless, this region offers a significant proportion of exportable quality crops. The area under grape cultivation has expanded by 9% per year during the last couple of decades [6]. Agriculture's energy requirements are mostly determined by two factors: To begin with, the type of farming technology used by the farmers, as well as the local weather conditions [7]. If the input energy is very high, it might be assumed that the producers are not making sufficient efforts to improve the production sys-

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tem's efficiency [8]. Agriculture operations like fuel and human labor are quite energy-intensive [9].

Nomenclature

AICTE	All India Council for Technical Education
NDF	National Doctoral Fellowship
Mt	Metric Tonnes
ha	Hectare
USD	US Dollar
NHB	National Horticulture Board
APEDA	Agricultural and Processed Food Products Export Development Authority
N-P-K	Nitrogen, Phosphorous, Potassium
P_2O_5	Phosphorus Pentoxide
K_2O	Potassium Oxide
ME	Machinery Energy
E	Energy equivalence (MJ/kg)
M	Mass of the machinery (kg)
T	Machine's total economic life (h)

2. Materials and methods

This research is carried out in the Sangli district, which is one of the major producers of grapes in Maharashtra. Survey data from various government organizations such as the National Horticulture Board (NHB) in Gurugram, India [3], and the Agricultural and Processed Food Products Export Development Authority (APEDA) in India are used to analyze the grape output. The NHB governs thorough descriptions, as well as onsite data from large grape growers in the regions. According to a survey done by NHB between 2012 and 2017, of the total grape production, Sangli has a share of 32% and 26% on Maharashtra and India levels respectively, making it a favorable location for this research [11]. To proceed with this energy analysis, the design model for this energy analysis is incorporated by taking into account the following factors: Data collected from farmers through questionnaires provided the input needed for energy analysis.

- The energy analysis execution section included taking into account the energy equivalence factors and formulae used to calculate the energy equivalent values for each source of energy.
- The output is presented as a percentage share of each type of energy considered, with the results displayed as a pie chart.

The energy inputs and outputs for all processes involved in grape production are calculated in this work. Energy analysis is split into two sections: data collecting and data interpretation as discussed below.

2.1. Data collection

The data was collected mainly from the statistical report published by NHB and also onsite data from major grape farms in the region. To perform the energy analysis, the data is inspected and gathered from major farms (10) from the Tasgaon taluka, Sangli, Maharashtra. These farms were chosen based on the productivity and quality of the grapes, accessibility, the likelihood of farmer collaboration, and the potential of data gathering. A schematic of the life cycle boundary for the energy analysis performed is shown in Fig. 1.

An intricate questionnaire (which includes a pre-defined set of questions) is used to collect data, and they were filled out on the spot after discussing and asking the corresponding questions with the farmers. Consequently, these questionnaires are prepared and filled which enables the examination of the various quantities of energy consumption. All of these farms produce a variety of high-quality grapes, with average yields ranging from 8 to 11 Mt per season per acre. The questionnaire included questions about the quantities of primary inputs used by the farms during the grape-growing process, as well as the number of grapes produced.

The data collected from both NHB and farms were used to calculate per-hectare input and output quantities for farms in the region. Depending on the sort of input being evaluated, the units of input varied. For human labor, the time spent performing a specific task was used as the unit, and the time spent per unit hectare was calculated using the data gathered.

2.2. Data interpretation

The interpretation of data is the second step in the energy analysis process. The total energy obtained in a single period of production based on data collected from the farms is used to perform the energy analysis of the grape production process. This is done by determining the total energy input for the total area used for grape growing in the region us-

Table 1
State-wise production of grapes from 2015-16 to 2017-18 [3]

Districts	2015-16		2016-17	
	Area (× 1000 Ha)	Production (× 1000 Mt)	Area (× 1000 Ha)	Production (× 1000 Mt)
Nashik	56.23	1237.06	56.27	1257.07
Sangli	21.45	611.27	21.68	617.74
Solapur	5.06	61.03	18.13	362.50
Pune	2.50	56.54	2.58	51.50

Table 2
Major grape-producing districts in Maharashtra [3,10]

Districts	2015-16		2016-17		2017-18	
	Area(× 1000 Ha)	Production (× 1000 Mt)	Area(× 1000 Ha)	Production (× 1000 Mt)	Area(× 1000 Ha)	Production (× 1000 Mt)
Maharashtra	90.09	2048.11	104.08	2378.17	105.50	2286.44
Karnataka	23.35	429.78	24.29	450.79	26.61	524.20
Tamil Nadu	2.44	34.10	2.49	29.02	2.16	58.93
Mizoram	2.47	22.55	2.45	18.00	2.45	18.00
Andhra Pradesh	0.73	14.64	0.77	15.36	0.80	15.92
Punjab	0.30	8.49	0.29	8.22	0.29	8.23
Telangana	0.59	13.54	0.29	3.47	0.35	5.27
Madhya Pradesh	0.16	2.20	0.09	1.28	0.09	1.28
Jammu & Kashmir	0.13	0.32	0.32	0.65	0.33	0.89
Nagaland	0.26	0.50	0.26	0.60	0.22	0.54
Kerala	1.00	15.50	1.20	16.00	0.01	0.18
Haryana	0.04	0.16	0.04	0.04	0.04	0.09
Total	121.65	2590.04	136.64	2921.70	138.91	2920.09

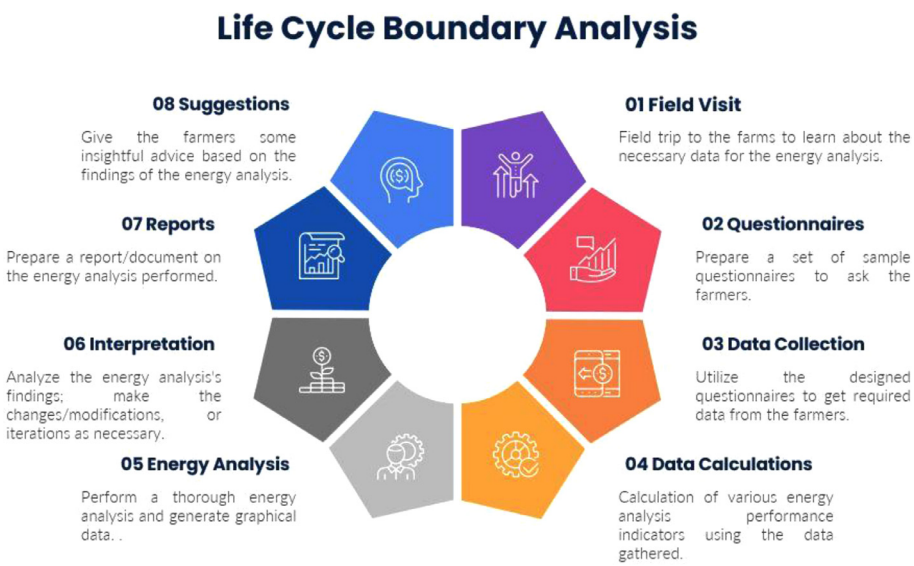

Fig. 1. A schematic of the life cycle boundary for the energy analysis performed.

Table 3
Energy parameters considered for performing energy analysis.

Parameter	Unit	Description
Indirect Energy	—	Indirect energy refers to the energy used to prepare or generate an input entity that is later used in the agricultural process.
Diesel	L	Diesel is utilized as a tractor fuel, which is required for tasks such as soil preparation, seed sowing, and other farm-related tasks.
Electricity	kWh	The water pump, as well as lighting in the tent house, paths, and other odd uses, all requires electricity.
Fertilizer	kg	To improve the nutrient content of the soil used for farming, fertilizer (rich in N-P-K) is necessary. Nitrogen, phosphorus pentoxide (P_2O_5), and potassium oxide (K_2O) are used.
Pesticide	kg	Pesticides are typically sprayed on plants to prevent bacteria and other contaminants from settling. Pesticides of many types, such as herbicides, insecticides, and fungicides, are employed.
Farm Machinery	hr	Tractors, spraying machines, simple blower machines, and water pumps are types of farm machinery. All of these farm machines are necessary to carry out various farming tasks such as soil preparation, seed sowing, fertilizer and pesticide spraying, transportation of commodities, irrigation, and so on.
Direct Energy	—	Direct energy inputs are the energy inputs used on the farm during the grape production process.
Human Labor	hr	During all stages of the grape production process, including pre-harvesting, harvesting, and post-harvesting, human labor is necessary to undertake hands-on work. The amount of male and female laborers hired on a paid basis to conduct agricultural work is referred to as human labor. In the majority of cases, the male-to-female ratio is kept equal.

ing the energy equivalency values for particular input per unit hectare area. The reason for this data being collected in a single period of production is because grape harvesting normally takes place from February to May (4 months). Furthermore, it is the peak period for grape harvesting. During this time, we obtained all relevant parameters for energy analysis, such as the amount of human work engaged; the amount of fuel consumed, and so on. Instead of collecting data at multiple places or over different periods, all of this information is gathered at the same time from all major farms during a single production period to provide a platform for comparison throughout the energy analysis [12,13]. Energy input consists of gross energy and other inputs of material or any source until it is used in the field [14]. Direct and indirect energy inputs are the two basic categories of energy inputs. All of the energy inputs used during the grape production process on the farm are classified as direct energy inputs. These include human labor, irrigation, and transportation, among other things. All the energy used previously to prepare or produce an input entity that is later used in the agricultural process is referred to as indirect energy inputs. Fertilizers, machines, electricity, and other items are among them [15–17]. Table 3 summarizes the significance of all the energy parameters considered for the energy analysis. The energy required to use the farm machinery is calculated as:

$$ME = \frac{E \times M}{T} (MJ/h) \tag{1}$$

Where ME is the energy used by the machinery for its usage and maintenance (MJ/h), M is the mass of the machinery being implemented (kg), E is the energy equivalence factor (62.7 MJ/kg) [4] and T is the machine's total economic life that is being used (h). The average tractor mass (M) is considered to be 2300 kg (This value is considered for calculation after consulting with the farmers). In the grapes production process, all energy equivalence factors are used to calculate the energy utilization of a specific input or output entity per unit hectare. The overall energy utilization for the region is calculated using data from farmers on the area used for grapes production and is tabulated in Table 4. The formulae used for performing energy analysis are tabulated in Table 5. As per the calculations for the input and output energy, the following performance parameters are estimated to provide a better understanding of energy usage for grape production in the region.

2.2.1. Energy ratio

The energy ratio is a metric that shows how input and output energy are related. It's the ratio of total output energy to total input energy. It provides us with a fundamental understanding of energy usage processes. A value larger than one is always preferred because it indicates that less energy was consumed in the grape production process. This entails identifying processes that consume more energy than is required; thereby applying effective solutions to reduce those energy consumptions [18].

$$\text{Energy ratio} = \frac{\text{Energy Output (MJ)}}{\text{Energy Input (MJ)}} \tag{2}$$

Table 4
Energy factors used in energy calculations.

Energy type	Unit	Energy Equivalent (MJ/Unit)	Source
Diesel	L	56.31	[4]
Electricity	kWh	11.93	[4]
Fertilizers: Nitrogen	kg	60	[4]
P_2O_5	kg	11.1	[4]
K_2O	kg	6.7	[4]
Farmyard manure	kg	0.3	[4]
Machinery: Tractor	hr	13.11	[4]
Spraying machine	hr	0.298	—
Simple blower	hr	0.043	—
Pump	hr	0.057	—
Human labor	man-hour	1.96	[4]
Pesticide: Herbicide	kg	254	[22]
Insecticide	kg	185	[22]
Fungicide	kg	107	[22]
Irrigation	m^3	0.63	[22]
Output (grapes)	kg	11.8	[23]

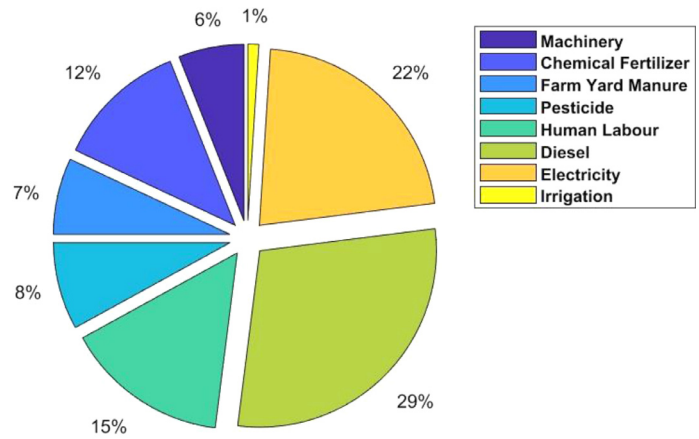

Fig. 2. Percentage distribution of various energy parameters

2.2.2. *Productivity per unit of energy*

The metric that depicts the relationship between grape production output and energy input is productivity per unit of energy. It's the ratio of grape production output (kg) to energy input (MJ). This value represents the grape's output per MJ of energy consumed. We must be able to identify the increase or decrease in grape production quantity for the same amount of energy used throughout a production period. It indirectly contributes to the quality of the methods used in grape production [19].

$$\text{Productivity/unit energy} = \frac{\text{Grapes Output (kg)}}{\text{Energy Input (MJ)}} \tag{3}$$

2.2.3. *Specific energy*

The ratio of energy input (MJ) to grape production output is known as specific energy (kg). It's just the inverse of productivity per unit of energy. It gives insight into the amount of energy used per kilogram of grapes produced. The grapes production process is optimal and energy-efficient when the value of specific energy consumption is low [20,21].

$$\text{Specific energy} = \frac{\text{Energy Input (MJ)}}{\text{Grapes Output (kg)}} \tag{4}$$

3. Results and discussions

This section gives a brief insight into the amount of energy equivalence for input and output energies required for the production of grapes for the Sangli.

The quantification of energy equivalent values (% and MJ) allows for the identification of areas with significant energy consumption. As a result, it directs the requirement for its minimizations by making the best use of the governing parameters. It also implies that the overall design model should be improved. Table 6 shows various quantities and energy requirements for several processes required for the same. It includes the farming process as input, quantities per unit area, and its associated energy equivalent values.

According to the results of the energy analysis, the total energy input required for the production of grapes is 39285.21 MJ, as indicated in Table 7, which reveals that the energy ratio for the Sangli district is 5.92 (approximately 6), indicating that the output energy produced is almost six times the input energy supplied. This value is validated from [23], which shows similar results (energy ratio = 6) in the line. A productivity ratio of 0.56 kg/MJ is obtained from the energy analysis. As a result, we can deduce that a productivity ratio may remain within a similar range for different geographic areas. The energy analysis calculates a specific ratio of 1.77 MJ/kg. The energy and productivity ratios should be as high as possible at all times. The specific ratio value should always be on the lower side.

The results of the energy analysis imply that diesel and electricity contribute 28.97% and 21.56% of total input energy, respectively as shown in Fig. 2. In studies carried out previously on grapes in Manisa province of Turkey, by [24], similar results were reported, with diesel and electricity having major shares of 33.04% and 21.86% of the input energy respectively. Similarly, in the research from the Antalya region of Turkey [23], the above values are 16.08% and 27.64% respectively. Thus, after comparing our obtained energy analysis values to that of the above-stated research works, the values for diesel and electricity input energy shares obtained from energy analysis are in lying in a similar range. The largest share is for diesel and electricity could be due to an overreliance on tractors and the use of electric pumps to draw water for irrigation [25]. Additionally, in several types of research, the energy ratio of grape production ranges from 6 to 8 [23,24,26], which is also in affirmation with the calculated energy ratio (5.92) from energy analysis.

4. Conclusions

This research has conducted a complete, systematic, and comprehensive energy analysis of the grape-growing process in the Sangli district of Maharashtra, India, highlighting the energy utilization in key farming activities. Energy analysis is a tool used to quantify the various energies involved in the production of grapes. It allows for the analysis of the

Table 5
Formulae used for unit consumption per unit area basis.

Parameter	Unit	Formula	
Machinery	(h/ha)	$\frac{\text{No. of machines} \times \text{No. of active working days} \times \text{No. of daily active working hours (h)}}{\text{Area (ha)}}$	(5)
Chemical Fertilizer & Farm Yard Manure	(kg/ha)	$\frac{\text{Quantity of chemical fertilizer used or Farm yard manure used for the last season (kg)}}{\text{Area (ha)}}$	(6)
Pesticide	(kg/ha)	$\frac{\text{Quantity of chemical pesticide used for the last season (kg)}}{\text{Area (ha)}}$	(7)
Human Labour	(h/ha)	$\frac{\text{Total no. of labours} \times \text{No. of active working days} \times \text{No. of daily active working hours (h)}}{\text{Area (ha)}}$	(8)
Diesel	(l/ha)	$\frac{\text{Volume of diesel used per day for farm machinery (L)} \times \text{No. of active working days}}{\text{Area (ha)}}$	(9)
Electricity	(kWh/ha)	$\frac{\text{No. of units consumed during the last season (kWh)}}{\text{Area (ha)}}$	(10)
Irrigation	(m^3/ha)	$\frac{\text{Volume of water used for irrigation per day}(m^3) \times \text{No. of days in the last farming period}}{\text{Area (ha)}}$	(11)

Table 6
Equivalent energy inputs and outputs for grapes production.

Inputs	Quantity per unit area (unit/acre)	Quantity per unit area (unit/ha)	Energy Equivalent (MJ)	Energy Equivalent (%)
Machinery (h)				
Tractor - land preparation	15.5	38.27	406.41	
Tractor-cultural practices	62	153.09	1625.64	
Tractor - harvesting	15.5	38.27	406.41	
Spraying machine	9	22.22	5.364	
Simple blower	9	22.22	0.774	
Pump	116	286.42	13.224	
Total	227	560.49	2457.82	6.26
Chemical fertilizer (kg)	30.86	76.20	4801.82	12.22
Farmyard manure (kg)	4833.33	11934.15	2899.99	7.38
Pesticide (kg)				
Herbicide	2.4	5.93	1219.2	
Insecticide	2.4	5.93	888	
Fungicide	4.77	11.78	1020.78	
Total	9.57	23.63	3127.98	7.96
Human labor (h)				
Land preparation	242.66	599.16	951.23	
Cultural practices	970.66	2396.69	3804.99	
Harvesting	242.66	599.16	951.23	
Total	1455.98	3595.01	5707.44	14.53
Diesel (L)	101.6	249.53	11381.38	28.97
Electricity (kWh)	355	876.54	8470.3	21.56
Irrigation (m^3)	348	859.26	438.48	1.12
Output (kg)	9850	24320.99	232460	100.00

Table 7
Energy input-output analysis.

Parameter	Unit	Value
Total energy input	MJ	39285.21
Energy ratio	—	5.92
Productivity ratio	kg/MJ	0.56
Specific ratio	MJ/kg	1.77

share and distribution of various sources of energy. Direct visits to major farms and discussions with the farmers as well as NHB data provided the inputs needed to do the energy calculations. The supplementary files included with this publication provide information about data collecting, the survey form, and other relevant fields. This research's key findings can be summed up as follows:

- The farmers in the Sangli district of Maharashtra are the respondents of an actual survey that provided the information needed for this research project's energy analysis.
- The farmers participate in a thorough survey to get all the data, facts, statistics, etc. required to understand the processes involved in producing raisins and grapes.
- An energy analysis is conducted for each case study involving a specific farmer, and the results are interpreted appropriately. As a result, the outcomes are displayed and given as a cumulative output for all the farmers' case studies taken into account.
- The total energy input required for the production of grapes is calculated to be 39285.21 MJ. Diesel (29%), electricity (22%), human labor (15%), chemical fertilizer (12%), pesticide (8%), farm yard manure (7%), machinery (6%), and irrigation (1%) are listed in decreasing order of energy usage.
- The energy ratio and productivity ratios are calculated as a result of the energy analysis; the corresponding values are 5.92 and 0.56 kg/MJ, respectively. The performance of the farm's output and returns are measured using energy analysis.
- When compared to the given input energy, the energy ratio signifies that almost six times more energy is obtained for the given input. A value of 0.56 kg/MJ obtained implies that 0.56 kg of grapes are produced for every MJ of energy consumed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.nexus.2022.100145.

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